

**Kathmandu University School of Education
Department of STEAM Education**

**Bachelor of Technical Education (B Tech Ed) Program
Information Technology (IT)**

In Association with the School of Engineering, KU

Course Curriculum

Semester VI

2024

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Overview of the B Tech Ed Program

Technical and Vocational Education and Training (TVET) has been one of the prominent sectors globally for a long. Dewey (1916) saw TVET as a tool for education reform in modernizing society. This has been equally vibrant in the present context as scholars (e.g., Maclean & Wilson, 2009) argue that TVET is an education and training that prepares an individual for gainful employment. TVET programs equip young people with the skills, knowledge, and competencies required to enter a particular type of professional career (OECD 2017). Hence, the role of TVET in preparing skilled human resources and enabling them to transition into the career path for gainful employment in specific and the country's economic development in a broad is inevitable. The formal TVET began in Nepal only around mid of the 20th century. Nevertheless, occupational skills have been transformed from generation to generation for a long (Ministry of Education [MoE], 2012). The establishment of the Council for Technical Education and Vocational Training (CTEVT), the governing body of TVET in 1989, was a substantive effort toward its development and expansion (CTEVT, 2019). Additionally, at present, the Centre for Education and Human Resource Development (CEHRD) offering technical diplomas in 284 technical stream schools throughout the country (GoN, 2018) and universities (such as Kathmandu University, Manmohan Technical University, etc.) run bachelor and master in TVET and other 11 federal ministries offer TVET related formal and non-formal programs (MoEST, 2019). The CTEVT, which is in charge of formulating TVET-relevant policies, developing standards for programs and curriculums, coordination, accreditation, monitoring, and supervision (CTEVT, 2019) largely shares the TVET related activities. There are 1131 CTEVT affiliated and constituent institutions with roughly the average capacity of 70,000 per annual operate formal TVET programs (CTEVT, 2020). Further expansion of TVET institutions in the country is yet to be expected soon since the government has a provision of establishing at least one technical school in all local levels.

Bachelor of Technical Education is a program to develop instructors and teachers in the field of Technical and Vocational Education and Training (TVET) with a view to fulfil the gap of acute shortage of instructors and TEVT professionals in the country which aspires to promote technical and vocational education two-third of the educational system and processes. The program has envisaged that a mix of two or more of the work-based learning, activity-based instruction, on the job training, project-based learning and problem-based learning be used to deliver the program. Because of the 'practical focus' of the program, the assessment has been proposed to address the needs of experiential learning.

The 136-credit program has been structured four types of the courses, such as Educational Core (18), Vocational Specialization (80), Vocational Pedagogy (20), and Educational Research and Internship (18) with a view to inculcate both pedagogical and vocational skills. The vocational specializations have been offered in a manner that they can be utilize hands-on activities as well. The program begins with the foundational courses in general education, vocational pedagogy and the area of specialization. In the later part of the program, OJT/WBL and forms of experiential learning shall be introduced. The culmination of the program will be internships in their specializations and a research report that addresses the problem of vocational education.

B Tech Ed in Information Technology

Bachelor of Technical Education is a 4-year (8 semesters) program to develop instructors, teachers, professionals in the field of Technical and Vocational Education and Training (TVET) with a view to fulfil the gap of acute shortage of instructors and TEVT professionals in the country which aspires to promote technical and vocational education two-third of the educational system and processes. The program has envisaged that a mix of two or more of the work-based learning, activity-based instruction, on the job training, project-based learning and problem-based learning be used to deliver the program. Because of the ‘practical focus’ of the program, the assessment has been proposed to address the needs of experiential learning. The B Tech Ed in IT program aims at providing technical, vocational, and educational exposures for students who can become IT professionals, IT Instructors, TVET educators, TVET curriculum developers, etc. The occupational/specialization courses emphasize the development of core IT skills in the field of computer science and technology innovation. The B Tech Ed in IT graduates will be eligible for doing master's program in IT, TVET, Education, Social Sciences, etc.

Major Objectives of the B Tech Ed in IT Program

- Demonstrate Level 6 competencies envisaged by the National Vocational Qualification Framework (NVQF) of Nepal,
- Exhibit comprehensive, meaningful, and coherent knowledge and skills in the field of technology and computer science
- Apply occupational specialization knowledge and skills in solving problems that occur in their contexts of work,
- Apply vocational pedagogical knowledge and skills in technical and vocational instruction of Information Technology,
- Show scholarly literacy, communication, quantitative reasoning, critical thinking, and learning skills needed for advanced and work-based learning,
- Exemplify a deep and principled understanding of the technical and vocational learning processes and the role of the instructors in facilitating these processes in the students.
- Apply a wide range of teaching process skills, including curriculum development, lesson planning, materials development, assessment, and pedagogical approaches, and
- Enable students to become resilient and lifelong learners by harnessing internal and external assets as TVET professionals.

Career Prospects

- TVET Instructors and Educators
- TVET Curriculum Developers
- IT Professionals (Software and Apps developer, programmer, Network Engineer, Web Developer, IT Admin, IT Engineer, etc.)
- Eligible for doing a master's program in IT, TVET, Education, Social Sciences, etc.

Entry Requirements

The minimum prerequisite for this program is the completion of Level 5 education (as per the National Vocational Qualification Framework) in relevant vocational specializations. The details are-

- Minimum 1.6 CGPA in 10+2
- Or
- A minimum 40% marks in 10+2, PCL or equivalent in any disciplines

Courses Offered in Semester VI

Subject	Credit Hour		
	Theory	Practical	Total
EDUC 301 Measurement and Evaluation	2	1	3
VPED 302 Training Methodology (2)	1	1	2
ITEX 310 Cybersecurity	1	1	2
ITEX 322 Cloud Computing	2	1	3
ITEX 323 Internet of Things (IoT), Concepts and Applications	1	1	2
ITEX 324 Artificial Intelligence	1	1	2
ITEX 325 Entrepreneurship	1	1	2
ITEX 326 IT and the Environment	1	1	2
Total	10	8	18

Core Courses

EDUC 301 Measurement and Evaluation

Course Name: Measurement and Evaluation

Code: EDUC 301

Credit Hour: 3 (2+1)

Nature: Theory + Practical

Course Description and Goals

This course, designed for Bachelor of Technical Education students, provides a comprehensive foundation in the principles, methods, and applications of educational measurement and evaluation. Through a blend of theoretical concepts and practical applications, students will explore the distinctions between tests, measurement, and evaluation and analyze the importance of reliability and validity in constructing effective assessments. The course will guide students in developing both objective and subjective test items, as well as using item analysis techniques to refine these items for enhanced reliability and validity. In addition, students will gain hands-on experience with formative assessment strategies, including rubrics, self and peer assessment, portfolios, and performance-based tasks. Finally, the course introduces technology-based assessment tools and techniques, enabling students to design and administer digital assessments, employ adaptive assessments, and explore virtual proctoring for secure online exams. By the end of this course, students will be equipped with the skills to apply assessment and evaluation concepts effectively, supporting quality educational practices and fostering meaningful learning outcomes.

Learning Outcomes

The following are the learning outcomes that students will develop:

- Interpret and differentiate core concepts in educational assessment, including test, measurement, evaluation, and assessment, and apply these distinctions to effectively support diverse educational objectives and student learning needs.
- Evaluate and apply key principles of reliability and validity in the construction and analysis of assessments, using item analysis techniques to enhance the accuracy and consistency of test items.
- Develop both objective and subjective assessment items, including multiple-choice, essay, and open-ended questions, ensuring alignment with instructional goals, reliability, and validity standards.
- Utilize a variety of formative and summative assessment strategies, such as rubrics, self-assessment, peer assessment, and performance-based tasks, to provide meaningful feedback that supports student growth and achievement.
- Integrate alternative assessment models like continuous assessment, portfolios, and performance-based tasks into instructional practices, fostering a holistic approach to evaluating student learning outcomes.
- Apply technology-enhanced assessment tools and techniques, including digital portfolios, online quizzes, adaptive assessments, and virtual proctoring, to create secure, personalized, and accessible assessment experiences in both physical and virtual learning environments.
- Design and evaluate instructional goals and objectives using frameworks such as Bloom's Taxonomy, ensuring that assessment practices are effectively aligned with desired learning outcomes and promote comprehensive understanding.

Content with specific objectives

Specific objectives	Contents
• Define, exemplify, and differentiate the concepts of Test, Measurement, and Evaluation • Differentiate and evaluate the ideas of Evaluation and Assessment • Define, apply, and differentiate the ideas of criterion and norm-reference tests • Evaluate, differentiate, and exemplify the concepts of summative and formative assessment • Describe and differentiate the concepts of assessment of, for, and as learning • Apply various levels of Bloom's taxonomy to create several instructional goals and objectives.	Module I: Overview of Measurement and Evaluation/Assessment (8 Hours) • Meaning and concept of Test, Measurement, and Evaluation • Difference between Measurement, Evaluation, and Assessment • Criterion-referenced and Norm-referenced tests • Difference between Evaluation and Assessment • Concept, Need, and Types of Assessment (Summative Vs. Formative) • Assessment of, for, as learning • Instructional Goals and Objectives (Revision of Bloom's Taxonomy)
• Define and evaluate the concepts of Reliability and Validity in educational assessment/evaluation • Define, exemplify, and differentiate the ideas/concepts of various types of reliability (mentioned in the contents) • Use different types of reliability in test items or test paper to make them effective • Interpret with examples various factors affecting reliability • Define and assess the concepts of validity in educational assessment/evaluation • Define, illustrate, and assess various types of validity • Apply various types of validity in the construction of test items	Module II: Reliability and Validity (8 Hours) • Reliability ○ Concept and meanings of Reliability ○ Types of Reliability (Test-Retest, Inter-rater, Parallel-forms, Internal Consistency) ○ Factors affecting Reliability • Validity ○ Concept and meaning of Validity in evaluation ○ Types of Validity (Content Validity, Construct Validity, Criterion Validity) ○ Factors affecting reliability

• Differentiate, with examples, between teacher-made and standardized tests • Construct various objective types of test items by ensuring reliability and validity • Build/develop various types of subjective items by ensuring reliability and validity • Define, differentiate, and compare and contrast the concepts: Item difficulty index, Discrimination Index, Distractor Analysis, and Internal Consistency • Use/Apply item analysis techniques to find various measures and refine the teacher-made tests • Discuss, with examples, various factors that influence test scores	Module III: Test Construction and Item Analysis (12 Hours) • Teacher-made Vs. Standardized Test • Construction of objective items (Multiple Choice Questions, True/False Items, Fill-in-the-Blank Items, Matching items, Assertion-Reason Questions) • Construction of Subjective Items (Short-answer, Essay, Long-answer, Problem-solving, Case Study, Reflective Questions, Open-ended questions) • Item Analysis Techniques: Item Difficulty Index, Discrimination Index, Distractor Analysis (for objective items), Internal Consistency (Cronbach Alpha) • Factor Influencing Test Scores (Nature of Test, Psychological Factors, and Environmental Factors)
• Describe and exemplify the concepts: Feedforward, feed up, and feedback, and differentiate them with examples • Interpret the meanings of Rubrics in educational assessment practices • Design Rubrics for some formative assessment tasks (Presentation, short/long/essay questions, Problem-solving, Group work, Reflective Writing, Classroom Participation, etc.) • Interpret and exemplify the concepts of Self and peer assessment, CAS, Portfolio, and performance-based assessment • Create some assessment tasks based on some popular alternative formative assessment models (Self and peer assessment, CAS, Portfolio, and performance-based assessment)	Module IV: Some Formative Assessment Techniques and Alternative Formative Assessments (10 Hours) • Concepts and applications of Feedforward, Feed up, and Feedback • Rubrics and Rubrics Development • Self-Assessment and Peer-Assessment • Continuous Assessment System (CAS) • Portfolio Assessment • Performance-based Assessment

• Evaluate the concepts of using Technology in summative and formative assessment methods • Interpret and analyze various technology-enhanced assessment methods (see contents) • Explore and use various technology-based assessment tools to create several assessment tools/tasks (e.g., quizzes) • Analyze the applications of technology in assessment practices • Explain and use some crucial virtual proctoring tools and exam security platforms for online examination purpose	Module V: Technology in Assessment (7 Hours) • Introduction to Technology in Assessment • Technology-enhanced Assessments (Online Quizzes & Exams, Automated Scoring System, Digital Portfolio, Game-based Assessment, Survey Response Tools) • Tools for creating and Administering Digital Assessment (including hands-on activity) • Formative Assessment with Technology (Technology for learning improvement) • Adaptive and Personalized Assessment • Virtual Proctoring and Exam Security
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Major Teaching and Learning Strategy

The following are the major teaching and learning strategies for facilitating the course contents effectively:

- Interactive Lecture and Discussions
- Presentation and Demonstration
- Case studies
- Technology integration
- Peer and group-based tasks

Practical and Project: Conduct at least two projects (Project 6 is compulsory)

1. Assessment Design Project

- **Objective:** Create a comprehensive assessment toolkit tailored for a specific subject area or grade level.
- **Tasks:**
 - Design a balanced mix of assessment types (formative, summative, and alternative assessments).
 - Include criterion-referenced and norm-referenced tests, objective and subjective items, and construct both traditional (e.g., multiple-choice) and creative formats (e.g., portfolios or case studies).
 - Develop rubrics for at least three different types of items to ensure alignment with instructional goals and Bloom's Taxonomy levels.
- **Outcome:** Submit a detailed toolkit with assessments, rubrics, and guidelines for administration and scoring.

2. Reliability and Validity Analysis

- **Objective:** Conduct a reliability and validity study for an assessment created by peers.
- **Tasks:**
 - Select an assessment created by another group and assess its reliability (e.g., test-retest, inter-rater).

- Analyze its validity by examining content, construct, and criterion validity.
- Identify factors that might affect the reliability and validity, suggesting improvements.
- **Outcome:** Present findings in a report, with data supporting reliability and validity levels and recommendations for refinement.

3. Technology-enhanced Formative Assessment

- **Objective:** Create a digital formative assessment using online tools.
- **Tasks:**
 - Use a platform like Google Forms, Kahoot, or Microsoft Forms to design a quiz that includes various question types (multiple choice, matching, open-ended).
 - Incorporate real-time feedback mechanisms (feedforward, feedback, and feed up) to guide learners.
 - Test the digital assessment with peers, gathering their feedback on its effectiveness and usability.
- **Outcome:** Submit a link to the digital assessment, along with a reflection report on the process and effectiveness.

4. Portfolio and Self-assessment System

- **Objective:** Design a portfolio assessment system that encourages self-assessment and peer assessment.
- **Tasks:**
 - Create a portfolio template that allows students to document their learning progress and reflections over time.
 - Design rubrics for self and peer assessments to be used at different stages of portfolio completion.
 - Implement the system with a peer, collecting artifacts and conducting both self-assessment and peer-assessment.
- **Outcome:** Present a sample portfolio and a summary report on how the self and peer assessment process influenced learning and assessment quality.

5. Adaptive Assessment Project

- **Objective:** Create an adaptive assessment framework using a simple decision tree or branching method.
- **Tasks:**
 - Design a series of questions that adapt based on student responses, ensuring differentiation in difficulty and content focus.
 - Use a tool like Google Forms with “go to section based on answer” features or create a paper-based branching model.
 - Test the adaptive assessment with peers, documenting how different pathways impact student learning and engagement.
- **Outcome:** Submit the adaptive assessment along with a report on the decision-making process and an analysis of peer feedback on the adaptive experience.

6. Item Analysis of a Teacher-Made Assessment

Objective: Conduct a comprehensive item analysis of a teacher-made assessment to evaluate its effectiveness using key metrics such as the difficulty index, discrimination index, and internal consistency.

Tasks:

- **Select an Assessment:**
 - Choose a recently administered teacher-made assessment (e.g., a quiz or test) from a specific subject area.
- **Data Collection:**
 - Gather the raw scores of all students who took the assessment. Ensure the dataset includes student identifiers for reference.
- **Calculate Item Difficulty Index:**
 - For each question on the assessment, calculate the difficulty index using the formula: $\text{Difficulty Index (p)} = \frac{\text{Number of Students Answering Correctly}}{\text{Total Number of Students}}$
 - Interpret the results to classify items as easy, moderate, or difficult.
- **Calculate Discrimination Index:**
 - Divide students into two groups: the upper and lower 27% based on their total test scores. For each item, calculate the discrimination index using the formula $\text{Discrimination Index (D)} = \frac{(\text{Upper group correct} - \text{Lower group correct})}{\text{Number of Students in each group}}$
 - Analyze the results to identify items that effectively discriminate between high and low performers.
- **Assess Internal Consistency:**
 - Calculate the internal consistency of the assessment using Cronbach's Alpha. Use a statistical software tool (e.g., SPSS, R, or Python) to compute this value. The formula is: $\alpha = \frac{N \cdot \bar{c}}{\bar{v} + (N-1) \cdot \bar{c}}$ where:
 - N = Number of items
 - $\bar{c}$ = Average covariance between item pairs
 - $\bar{v}$ = average variance of each item
 - Interpret the Cronbach's Alpha value to evaluate the reliability of the assessment.
- **Report Findings:**
 - Compile the results of the item analysis in a report that includes:
 - Summary statistics of difficulty and discrimination indices for each item.
 - A discussion of the implications of these indices on the assessment's effectiveness.
 - Recommendations for improving poorly performing items based on the analysis.
 - Summarizing the overall reliability of the assessment based on Cronbach's Alpha value.
- **Presentation:**
 - Present the findings to the class, including a visual representation (e.g., charts or graphs) of the difficulty and discrimination indices for better understanding.

Outcome: Students will gain hands-on experience with item analysis, understanding how to apply statistical methods to assess the quality of assessment items, and enhance their effectiveness in measuring student learning.

Assessment Plan

- **In-semester (50)**
Assignment, written exam, presentation, case study, report, objectives, attendance
- **End-semester (50)**

Question Types	Marks Distribution	Total Question
MCQ	10 Marks	10 Questions
SAQ	24 Marks	7 Questions [Attempt 6 Questions]
LAQ	16 Marks	3 Questions [Attempt 2 Questions]

Marking Scheme for Final Examination:

- Section A (**MCQ**): 10 Questions $\times$ 1 marks
- Section B (**Short Answer Question-SAQ**) – Total 7 Questions $\times$ 4 marks (Attempt any Six)
- Section C (**Long Answer Question-LAQ**) – Total 3 Questions $\times$ 8 marks (Attempt any Two)

Learning Units	Total Credit Hours	MCQs	SQs	LQs	Remarks
Module I: Overview of Measurement and Evaluation/Assessment	8 Hours	1	1		*Long questions can be asked from this unit, too.
Module II: Reliability and Validity	8 Hours	2	1	1	
Module III: Test Construction and Item Analysis	12 Hours	3	2	1	
Module IV: Some Formative Assessment Techniques and Alternative Formative Assessments	10 Hours	3	2	1	
Module V: Technology in Assessment	7 Hours	1	1		
Total Questions	45 Hours	10	7	3	

References:

- Kubiszyn, T., & Borich, G. D. (2013). *Educational testing and measurement: Classroom application and practice* (10th ed.). Wiley.
- Brookhart, S. M. (2013). *How to create rubrics and use rubrics for formative assessment*. ASCD.

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- Almarode, J., Fisher, D., Frey, N. (2023). *How feedback works: A playbook*. Corwin.
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- Gardner, J. (Ed.)(2006). *Assessment and learning*. Sage Publications.

Vocational Pedagogical Course

VPED 302 Training Methodology

Course Title: Training Methodology

Course Code: VPED 302

Credit Hours: 2 (1 TH + 1 PR)

Nature: Theory + Practical

Course Description and goals

This course aims to provide students with the foundational knowledge and practical skills required for effective training methodology. The course's primary focus is to provide an in-depth understanding of the need assessment phase, including designing, delivering, and evaluating training programs. During the course, students will have the opportunity to delve into both theoretical and practical aspects. This will enable them to cultivate a comprehensive and well-rounded approach to training that is applicable to a wide range of professional environments. The course integrates learning strategies, instructional techniques, and monitoring tools to prepare learners for real-world training environments.

Learning Outcomes

- Conduct training needs assessment to identify gaps and opportunities.
- Design comprehensive training plans, including logistics and alternative approaches.
- Deliver training sessions using effective instructional and engagement strategies.
- Monitor and evaluate training programs using appropriate tools and models, such as the Krick Patrick's model.

Content with specific objectives

Specific objectives	Contents
• To define need assessment in a training context. • To explain the importance of identifying training needs. • To differentiate reactive and proactive need assessment. • To apply methods for conducting reactive and proactive assessments. • To create a need assessment report based on a simulation scenario.	1. Need Assessment in Training 4 hrs. 1.1. Concept of need assessment in the training context 1.2. Importance of identifying training needs 1.3. Reactive vs Proactive need assessment 1.4. Methods for conducting assessments 1.5. Simulation: Creating a need assessment report
• To define training design and identify its key elements. • To explain the significance of a well-structured training design. • To develop a training plan including schedule, venue setup, and resource management.	2. Training Desing 6 hrs. 2.1. Definition of training design 2.2. Developing a well-structured training design 2.2.1. <i>Aligning design with need assessment outcomes</i> 2.2.2. <i>Defining clear, measurable learning objectives</i> 2.2.3. <i>Structuring content</i> 2.3. Developing a training plan

To create contingency plans for training and manage logistics effectively.	2.3.1. <i>Setting a training schedule</i> 2.3.2. <i>Venue and resource management</i> 2.3.3. <i>Budgeting and logistic planning</i> 2.4. Contingency planning in training
- To discuss methods for delivering training effectively. - To design engagement activities to involve participants actively in the learning process. - To demonstrate key instructional techniques for clear and impactful training delivery.	3. Training Delivery 6 hrs. 3.1. Effective training delivery methods 3.2. Engagement activities for active learning 3.3. Instructional techniques for clear communication 3.3.1. <i>Presentation skills, pacing and tone</i> 3.3.2. <i>Handling Q&A sessions effectively</i> 3.3.3. <i>Providing feedback and reinforcement during training</i>
- To identify different types of training, monitoring, and evaluation strategies. - To apply Krick Patrick's Model to assess training effectiveness. - To utilize tools to monitor and evaluate training outcomes.	4. Training Monitoring and Evaluation 4 hrs. 4.1. Monitoring Training 4.2. Evaluation strategies 4.3. Kirkpatrick's Model of Training Evaluation 4.4. Tools for monitoring and evaluation
- To enable students to draft a comprehensive training plan. - To encourage students to deliver the mini-training sessions.	5. Practicum 10 hrs. <i>Students will organize and deliver the mini training by applying the content from the Training Methodology course.</i>

Major Teaching and Learning Strategy

- Hybrid-based teaching:** Present information to students through oral presentation and deliver instructional content outside of class through videos or PowerPoint.
- Inquiry-based and active learning:** Encourage students to ask questions and engage them to participate actively in the learning process. (For example: Group Discussion or Case Studies)
- Hands-on Practical Sessions:** Allow students to engage in fieldwork, design training materials, and deliver mock training sessions.
- Field Visits:** Engage in real-world assessments of training needs and delivery.

Practical [10 hrs.]

Students will organize and deliver the training by applying the content from the Training Methodology course in a step-by-step manner, including the development of a need assessment report, training design and delivery, monitoring, and evaluation of the training.

- ✓ Conduct a need assessment of a group of 4 to 5 using at least one method, such as surveys or interviews with the target audience.
- ✓ Draft a need assessment report outlining the identified training needs.
- ✓ Develop a training plan, with each member assigning specific roles.
- ✓ Organize and deliver mini training on a selected topic.
- ✓ Monitor and evaluate the training by applying Kirkpatrick's Model.
- ✓ Submit the final report summarizing the outcomes, insights, and areas of improvement from the training you have delivered.

Assessment Plan

- **In-semester (50)**
Assignment, written exam, presentation, case study, report, objectives, attendance
- **End-semester (50)**

Question Types	Marks Distribution	Total Question
MCQ	10 Marks	10 Questions
SAQ	24 Marks	7 Questions [Attempt 6 Questions]
LAQ	16 Marks	3 Questions [Attempt 2 Questions]

Marking Scheme for Final Examination:

- Section A (**MCQ**): 10 Questions × 1 marks
- Section B (**Short Answer Question-SAQ**) – Total 7 Questions × 4 marks (Attempt any Six)
- Section C (**Long Answer Question-LAQ**) – Total 3 Questions × 8 marks (Attempt any Two)

Learning Units	Total Credit Hours	MCQs	SQs	LQs	Remarks
Need Assessment in Training	4	2	1	0.5	
Training Designing	6	3	2	1	
Training Delivery	6	3	2	1	
Training Monitoring and Evaluation	4	2	2	0.5	
Practicum	10	0	0	0	

References:

- Barbazette, J. (2006). *Training Needs Assessment*. Pfeiffer.
- Lawson, K. (2016). *The Trainer's Handbook*. John Wiley & Sons.
- Piskurich, G. M. (2000). *The ASTD Handbook of Training Design and Delivery*. McGraw-Hill Profession.

ITEX 323 Internet of Things (IoT), Concepts and Applications

Course Name: ITEX 323 Internet of Things (IoT), Concepts and Applications **Code:** ITEX 323

Credit Hour: 2 (1 TH + 1 PR)

Nature: Theory + Practical

Course Description and goals

write in a paragraph.

The course "Internet of Things (IoT), Concepts and Applications" (ITEX 323) is designed to provide students with a comprehensive understanding of the fundamental concepts and technologies involved in the Internet of Things (IoT). Through a combination of theoretical knowledge and practical experience, students will explore various IoT applications and use cases across different domains. The course aims to equip students with the skills necessary to design and develop IoT-based systems, emphasizing real-world applications. This core course will prepare students to leverage IoT technologies in innovative ways, ultimately enhancing their technical expertise and expanding their professional opportunities.

Learning Outcomes

- Describe the basic architecture and components of IoT.
- Analyze different communication protocols used in IoT.
- Develop simple IoT applications using appropriate hardware and software tools.
- Evaluate the potential and challenges of IoT in various sectors.

Content with specific objectives

Specific objectives	Contents
• Define IoT and its key characteristics. • Describe the fundamental aspects of IoT.	Unit 1 - Introduction to IoT (3 Hours) • Definition and Characteristics of IoT • IoT Architectures (3-layer, 5-layer models) • IoT vs. M2M vs. CPS
• Identify key IoT communication technologies (e.g., RFID, NFC, Bluetooth, Zigbee, LoRaWAN). • Describe the functions and applications of different IoT communication technologies.	Unit 2 - IoT Communication Protocols (3 Hours) • Wireless Sensor Networks • RFID, NFC, Bluetooth, Zigbee, LoRaWAN IoT Network Protocols: MQTT, CoAP, HTTP/HTTPS
• Identify different IoT hardware components (e.g., sensors, actuators, microcontrollers like Arduino and Raspberry Pi). • Describe the functions and uses of various IoT hardware.	Unit 3 - IoT Devices and Platforms (3 Hours) • Sensors, Actuators, and Microcontrollers (Arduino, Raspberry Pi) • IoT Development Platforms (Google Cloud IoT, AWS IoT, IBM Watson IoT)

• Describe IoT data management processes including data collection, storage, and processing. • Analyze data management processes support IoT applications.	Unit 4 - Data Management and Analytics in IoT (3 Hours) • Data Collection, Storage, and Processing • Cloud Computing for IoT • Big Data Analytics in IoT
• Identify IoT applications across different industries (e.g., Smart Cities, Smart Homes, Healthcare, Industrial IoT). • Explain how IoT is utilized in various sectors and the benefits it provides.	Unit 5 - IoT Applications and Case Studies (3 Hours) • Smart Cities, Smart Homes, and Healthcare • Industrial IoT (IIoT) • Challenges in IoT (Security, Privacy, Scalability)

Detailed Course Plan

Unit	Lesson/Course Plan
1. Introduction to IoT (3 Hours)	• Defining IoT: Understanding the core concepts and components • Exploring IoT Applications: Real-world examples and use cases • Understanding IoT Architectures: Three-tier and five-tier models • Exploring IoT Communication Protocols: Wireless sensor networks, RFID, NFC, Bluetooth, Zigbee, LoRaWAN, MQTT, CoAP, HTTP/HTTPS • Sensors and Actuators: Types, functions, and applications • IoT Development Platforms: Cloud-based platforms (AWS IoT, Azure IoT, Google Cloud IoT) and on-device platforms (Arduino, Raspberry Pi)-----
2. IoT Communication Protocols (3 Hours)	• Define wireless sensor networks and their components • Discuss the challenges and advantages of WSNs in IoT • Explore popular WSN protocols (e.g., IEEE 802.15.4, ZigBee) • Compare and contrast different protocols based on range, power consumption, and data rates • Discuss real-world applications of WSNs (e.g., environmental monitoring, smart agriculture) • RFID Technology • Explain the principles of RFID and its components (tags, readers, antennas) • Discuss different RFID standards (e.g., RFID-HF, RFID-UHF) • Explain the concept of Near-Field Communication (NFC) • Discuss NFC applications (e.g., contactless payments, data exchange) • Compare RFID and NFC • Bluetooth and Zigbee

	• Discuss the features and applications of Bluetooth • Explore Bluetooth Low Energy (BLE) for IoT • Zigbee Technology • Explain the Zigbee protocol and its characteristics • Discuss Zigbee applications in IoT • LoRaWAN Overview • Introduce LoRaWAN as a long-range, low-power wireless protocol • Discuss the key features and benefits of LoRaWAN • LoRaWAN Architecture and Components • Explain the components of a LoRaWAN network (gateways, devices, network servers) • Discuss the communication process in LoRaWAN
3. IoT Devices and Platforms (3 Hours)	• Introduction to Sensors: • Types of sensors (temperature, humidity, light, motion, etc.) • Sensor selection criteria (accuracy, range, power consumption) • Actuators: • Types of actuators (motors, LEDs, relays) • Applications of actuators in IoT • Introduction to Microcontrollers: • Overview of microcontrollers (Arduino, Raspberry Pi, ESP32) • Key features and capabilities of microcontrollers • Programming Microcontrollers: • Introduction to programming languages for microcontrollers (C, C++, Python) • Basic programming concepts and examples • Cloud-Based Platforms: • Google Cloud IoT • AWS IoT • IBM Watson IoT • Features and Benefits: • Data ingestion, processing, and analysis • Device management and connectivity • Integration with other cloud services • Choosing the Right Platform: • Factors to consider when selecting an IoT platform (scalability, cost, features)
4. Data Management and Analytics in IoT (3 Hours)	• Methods of data collection in IoT (sensors, APIs, user-generated data) • Challenges of data collection in IoT (data quality, volume, velocity)

	• IoT data storage options (edge devices, cloud storage, databases) • Considerations for data storage (scalability, cost, reliability) • Data cleaning, preprocessing, and transformation • Real-time data processing and analytics • Role of Cloud Computing in IoT: • Benefits of using cloud platforms for IoT (scalability, cost-effectiveness, flexibility) • Cloud services for IoT (storage, compute, analytics) • Cloud Platforms for IoT: • Overview of popular cloud platforms (AWS IoT, Azure IoT, Google Cloud IoT) • Comparison of features and capabilities • Introduction to Big Data: • Characteristics of big data (volume, velocity, variety, veracity) • Challenges of big data analytics • Big Data Tools and Technologies: • Hadoop, Spark, NoSQL databases • Data visualization tools • Applications of Big Data in IoT: • Predictive maintenance, anomaly detection, customer insights
5. IoT Applications and Case Studies (3 Hours)	• Smart Cities: Overview, IoT applications in urban infrastructure (smart grids, transportation, waste management) • Smart Homes: IoT devices and systems for smart homes (smart thermostats, security systems, home automation), • Healthcare: IoT applications in healthcare (remote patient monitoring, wearable devices, medical devices) • Impact of IoT on healthcare delivery and outcomes • Introduction to IIoT: Definition and applications of IIoT • Benefits of IIoT for industries (increased efficiency, cost reduction, improved quality) • Security: • Privacy: • Scalability
<u>PRACTICAL SESSIONS</u>	
Session 1: Introduction to IoT Development Environment (3 Hours)	• Setting up IoT Development Platforms (e.g., Arduino IDE, Raspberry Pi) • Basic programming for IoT devices

Session 2: Working with Sensors and Actuators (3 Hours)	• Interfacing sensors (Temperature, Humidity, Proximity) • Controlling actuators (Motors, LEDs)
Session 3: IoT Communication Protocols (3 Hours)	• Implementing MQTT and HTTP protocols • Sending sensor data to the cloud
Session 4: Data Collection and Analysis (3 Hours)	• Collecting data from IoT devices • Storing data in cloud databases • Basic data analytics using Python
Session 5: IoT Project Development (3 Hours)	• Designing a small IoT application (e.g., Smart Home System) • Integrating sensors, communication, and cloud services • Presenting the project

Major Teaching and Learning Strategy (List of some Teaching and Learning Strategies for the faculty)

1. Project-Based Learning

- Group Projects: Assign students to work in groups on IoT projects.
- Problem-Based Learning: Present students with real-world challenges and guide them through the process of finding IoT-based solutions.

2. Case Studies and Discussions

- Industry Examples: Discuss real-world IoT applications and case studies.
- Class Discussions: Facilitate discussions on the ethical, social, and economic implications of IoT.

3. Online Resources and Tools

4. Simulated Environments

Assessment Plan

• **In-semester (50)**

Assignment, written exam, presentation, case study, report, objectives, attendance

• **End-semester (50)**

Question Types	Marks Distribution	Total Question
MCQ	10 Marks	10 Questions
SAQ	24 Marks	7 Questions [Attempt 6 Questions]
LAQ	16 Marks	3 Questions [Attempt 2 Questions]

Marking Scheme for Final Examination:

- Section A (MCQ): 10 Questions × 1 marks
- Section B (Short Answer Question-SAQ) – Total 7 Questions × 4 marks (Attempt any Six)

- Section C (**Long Answer Question-LAQ**) – Total 3 Questions × 8 marks
(Attempt any Two)

Learning Units	Total Credit Hours	MCQs	SQs	LQs	Remarks
Chapter 1	6	4	1	.5	
Chapter 2	6	4	1	.5	
Chapter 3	6	4	1	1	
Chapter 4	6	4	2	.5	
Chapter 5	6	4	2	.5	

References:

- **Primary Textbook:**
 - Bahga, A., & Madisetti, V. (2014). *Internet of Things: A Hands-On Approach*. VPT.
- **References:**
 - McEwen, A., & Cassimally, H. (2013). *Designing the Internet of Things*. Wiley.
 - Raj, P., & Raman, A. C. (2017). *The Internet of Things: Enabling Technologies, Platforms, and Use Cases*. CRC Press.
 - Minerva, R., Biru, A., & Rotondi, D. (2015). *Towards a Definition of the Internet of Things (IoT)*. IEEE Internet Initiative.

Occupational/Specialization Course

ITEX 322 Cloud Computing

Course Title: Cloud Computing

Course Code: ITEX 322

Credit Hours: 3 (2 TH +1 PR)

Nature of the course: Theory and Practical

Course Description and Goals

This course is introductory and aims to include Cloud computing features, challenges and its opportunities. This course will cover various aspects of cloud in terms of its deployment and service models, technology and challenges. The practical implementation of cloud-based services deployment, security, and scalability practices will be covered by the course.

Learning Outcomes

On completion of this course, the students will be able to:

- Cloud computing architecture and its benefits
- Understand the technologies involved in the cloud computing
- Familiarize with cloud application deployment and service model
- Differentiate between other distributed computing and cloud computing
- Practical hand-on session on different cloud platforms, application deployment, and security constraint
- Familiar with emerging tools and technology in cloud-based application

Content with specific objectives

Specific objectives	Contents
• To familiarize yourself with the concept of cloud computing • To discuss the features and challenges of cloud computing • To discuss state of the art of cloud computing	Unit I: Introduction [4 hours] 1.1. Introduction to Cloud Computing 1.2. The Evolution of Cloud Computing 1.3. State of Arts of Cloud Computing 1.4. Features of Cloud Computing 1.5. Challenges of Cloud Computing
• To discuss about various service models and its characteristics • To learn about the applicability of the models in different scenarios	Unit II: Cloud Service Delivery Model [4 hours] 2.1. Introduction to Types of Cloud Services 2.2. Software as a Service (SaaS) 2.3. Platform as a Service (PaaS) 2.4. Infrastructure as a Service (IaaS) 2.5. Other Services such as (DaaS, XaaS, etc)
• To discuss various deployment models and their characteristics • To learn about the applicability of the models in different scenarios	Unit III: Cloud Deployment Model [5 hours] 3.1. Introduction to Cloud Types 3.2. Private Cloud 3.3. Public Cloud 3.4. Hybrid Cloud 3.5. Community Cloud

● To learn about the technology background of cloud computing ● To discuss and differentiate the cloud from other distributed computing ● To explore different proprietary and open-source cloud software ● To discuss the challenge of the cloud from the perspective of security	Unit IV: Cloud Technologies [6 hours] 4.1. Introduction to Cloud Technology 4.2. Virtualization 4.3. Cluster Computing 4.4. Grid Computing 4.5. Distributed Computing 4.6. Cloud Platforms
● To discuss security challenges and their probable solutions in the cloud ● To explore different services and micro-service architecture	Unit V: Cloud Performance [8 hours] 5.1. Cloud Security 5.2. Service Oriented Architecture 5.3. Microservice Architecture 5.4. Cloud Native 5.5. Container and Orchestration 5.6. Availability and Reliability
	Unit VI: Cloud Application in Emerging Technology [3 hours] 6.1. IoT 6.2. Big Data

Major Teaching and Learning Strategy

- **Interactive class sessions**
 - The understanding of cloud and distributed computing will be discussed with the implementation of an active learning procedure. All students will be voluntarily involved in their understanding of bare metal servers, clusters, grids, and clouds.
 - The understanding will be further refined with the introduction and distinct characteristics of cloud computing and how it has transformed the service and software deployment and development.
- **Workshop on Cloud Platform**
 - Hands-on sessions on different proprietary cloud platforms. Students will explore the free tier services of AWS and AZURE in the workshop.
- **Group Presentation and Discussion**
 - Students will be divided into groups of four to five members. Group presentation on cloud applications, exploration, and service delivery from cloud platform. The presentation will include their understanding and knowledge of the course in the cloud platform, which may range from service delivery to analyzing cloud performance and its application in emerging technology.
- **Collaborative work and Field Visit**
 - Students attend a collaborative session with providers and practitioners of cloud service. The session could be online or, if applicable, a field visit. The session

will be helpful for students to become familiar with industry practices in cloud service or the experience of cloud practitioners.

Practical (15 Hours)

Exploration of cloud platforms

AWS, Azure and OpenStack (10 hours)

Familiarization with basic services provided by proprietary cloud service providers and open-source cloud service software.

Application deployment from cloud platforms along with different perspectives of performance and security.

Group Presentation: (5 hours)

Assessment Plan

- **In-semester (50)**
Assignment, written exam, presentation, case study, report, objectives, attendance
- **End-semester (50)**

Question Types	Marks Distribution	Total Question
MCQ	10 Marks	10 Questions
SAQ	24 Marks	7 Questions [Attempt 6 Questions]
LAQ	16 Marks	3 Questions [Attempt 2 Questions]

Marking Scheme for Final Examination:

- Section A (**MCQ**): 10 Questions $\times$ 1 marks
- Section B (**Short Answer Question-SAQ**) – Total 7 Questions $\times$ 4 marks
(Attempt any Six)
- Section C (**Long Answer Question-LAQ**) – Total 3 Questions $\times$ 8 marks
(Attempt any Two)

Learning Units	Total Credit Hours	MCQs	SQs	LQs	Remarks
1. Introduction	4	2	1	0	
2. Cloud Service Delivery Model	3	1	1	0.5	
3. Cloud Deployment Model	3	1	1	0	
4. Cloud Technology	5	2	1	1	
5. Cloud Performance	8	3	3	1.5	
6. Cloud Application in Emerging Technology	3	1			

References:

- Furht Borko, Handbook of Cloud Computing, Springer, 2010, First Edition
- Mather Tim, Kumarswamy Subra, Latif Sahid, Cloud Security and Privacy, O'Reilly, 2009, First Edition
- Ricardo Puttini, Thomas Erl, and Zaigham Mahmood, Cloud Computing: Concepts, Technology & Architecture,
- James F. Ransome, John W. Rittinghouse Cloud Computing Implementation Management and Security

ITEX 324 Artificial Intelligence

Course: Artificial Intelligence

Course Code: ITEX 324

Credit Hours: 2 (1 TH + 1 PR)

Nature: Theory and Practical

Course Description and Goals

This course provides an introduction to the basic concepts and techniques of Artificial Intelligence. Topics include search algorithms, knowledge, reasoning, machine learning, neural networks, and ethics in AI. Students will engage in both theoretical learning and practical applications, culminating in a project that applies AI techniques to solve real-world problems.

Learning Outcomes

On completion of this course, the students will be able to:

- Understand the basic principles and techniques of AI.
- Implement AI search algorithms to solve problems.
- Apply machine learning algorithms for classification, regression, and clustering tasks.
- Explore logic programming.
- Analyze ethical issues and the societal impact of AI technologies.
- Develop small AI projects to apply theoretical knowledge practically.

Content with specific objectives

Specific objectives	Contents
• Define Artificial Intelligence and its scope. • Discuss the history and evolution of AI. • Explore AI applications in various domains (healthcare, finance, robotics, etc.). • Understand the types of AI: Narrow, General, and Super AI.	Unit 1: Introduction to Artificial Intelligence (3 hrs) 1. Introduction to AI 2. History of AI 3. Types of AI 3.1. Narrow AI 3.2. General AI 3.3. Super AI 4. AI Applications
• Understand problem-solving agents • Learn about and implement uninformed and informed search algorithms	Unit 2: Problem Solving by Searching (6 hrs) 1. Problem-Solving Agents 2. Example Problems 3. Search Algorithms 4. Uninformed Search Strategies 4.1. Breadth-First Search (BFS) 4.2. Depth-First Search (DFS) 5. Informed Search Strategies 5.1. A* search 5.2. Greedy search
• Understand how AI systems represent knowledge using logic. • Explore propositional and predicate logic. • Understand how uncertainty can be quantified.	Unit 3: Knowledge and Reasoning (6 hrs) 1. Knowledge, Knowledge Base, Sentences, Knowledge Representation, Inference 2. Logic 2.1. Propositional Logic 2.1.1. Syntax 2.1.2. Semantics 2.1.3. Inference rules

	2.2. First-Order Logic 2.2.1. Models for First-Order Logic 2.2.2. Symbols and Interpretations 2.2.3. Terms, Atomic Sentences, Quantifiers 2.2.4. First-Order Inference 3. Uncertain Knowledge 3.1. Basic Probability Notation 3.2. Bayes' Rule and its Use 3.3. Naive Bayes Models
- Define machine learning and explain its types. - Learn about algorithms for classification and regression, and clustering, and implement some of them. - Understand the structure and function of neural networks.	Unit 3: Introduction to Machine Learning (12 hrs) 1. Types of Machine Learning 1.1. Supervised Learning 1.2. Unsupervised Learning 1.3. Semi-supervised Learning 1.4. Reinforcement Learning 2. Machine Learning Models 2.1. Decision Tree 2.2. k-Nearest Neighbors 2.3. Linear Regression 2.4. Logistic Regression 3. Clustering Algorithms 3.1. K-means Clustering 3.2. Hierarchical Clustering 4. Neural Networks 4.1. Feedforward Network 4.2. Activation Functions 4.3. Back-Propagation Algorithm
- Discuss ethical concerns related to AI. - Explore the societal impacts of AI in areas such as privacy, employment, and decision-making. - Understand the importance of responsible AI development.	Unit 4: Ethics, and Safety of AI (3 hrs) 1. Risks and Benefits of AI 2. The Limits of AI 3. The Ethics of AI

Major Teaching and Learning Strategy

- Interactive Lectures with Visualization:** Use interactive teaching tools like diagrams, flowcharts, and visual representations to explain AI algorithms, models, and processes.
- Problem-based Learning:** Engage students with real-world problems that require AI techniques to solve.
- Hands-on Labs:** Implement lab sessions after each theoretical lecture to reinforce learning through practical exercises.
- Project-based Learning / Active Learning:** Assign mini-projects that integrate AI topics covered in class, encouraging creativity and exploration.
- Case studies and discussions:** Incorporate case studies and discussions on real-world applications of AI and its ethical implications, encouraging critical thinking.

Practical / Lab Works

Some examples of lab works or practicals that can be carried out for this course are listed below:

1. Implement BFS and DFS to solve the 8-puzzle problem or maze navigation.
2. Implement linear regression to predict housing prices.
3. Implement the K-Nearest Neighbors algorithm to classify the Iris dataset.
4. Implement the K-means algorithm to find clusters on a dataset and visualize the clusters.
5. Solve logical puzzles using Prolog or LISP.

Assessment Plan

- **In-semester (50)**

Assignment, written exam, presentation, case study, report, objectives, attendance

- **End-semester (50)**

Question Types	Marks Distribution	Total Question
MCQ	10 Marks	10 Questions
SAQ	24 Marks	7 Questions [Attempt 6 Questions]
LAQ	16 Marks	3 Questions [Attempt 2 Questions]

Marking Scheme for Final Examination:

- Section A (MCQ): 10 Questions $\times$ 1 marks
- Section B (Short Answer Question-SAQ) – Total 7 Questions $\times$ 4 marks (Attempt any Six)
- Section C (Long Answer Question-LAQ) – Total 3 Questions $\times$ 8 marks (Attempt any Two)

Learning Units	Total Credit Hours	MCQs	SQs	LQs	Remarks
Unit 1	3	1	1		
Unit 2	6	2	1-2	1	
Unit 3	6	2	1-2	1	
Unit 4	12	4	2-3	1	
Unit 5	3	1	1		
Total	30	10	7	3	

Text Book:

“Artificial Intelligence: A Modern Approach” by Stuart Russell and Peter Norvig

Reference Books:

- “Machine Learning” by Tom M. Mitchell
- “Artificial Intelligence” by Elaine Rich, Kevin Knight, Shivashankar B. Nair
- “Artificial Intelligence: Structures and Strategies for Complex Problem Solving” by George F. Luger

ITEX 310 Cyber Security

Course Title: Cyber Security

Course Code: ITEX 310

Credit Hours: 2 (1 TH + 1 PR)

Nature: Theory +Practical

Course Description and Goals

This course provides a foundational understanding of cyber security, covering network, operating system, and web security essentials. Students will explore threat analysis, secure coding, cloud security, and incident response planning while gaining hands-on experience with tools like firewalls, IDS/IPS, and ZAP. Emphasizing continuous learning, the course prepares students to adapt to emerging cyber security trends, foster a security-aware culture, and apply best practices across diverse digital environments.

Learning Outcomes

On completion of this course, the students will be able to:

- Define and critically assess key cyber security principles, including risk management, encryption, and the CIA Triad (Confidentiality, Integrity, Availability)
- Identify various cyber threats (e.g., malware, phishing, ransomware) and apply appropriate defense strategies
- Implement network and operating system security practices, such as firewall configuration, patch management, and antivirus protection
- Identify common web application vulnerabilities and implement best practices for securing web applications, including secure coding and HTTPS
- Evaluate and adapt to emerging cyber security trends, such as cloud security, threat intelligence, and advanced technologies

Content with specific objectives

Specific objectives	Contents
• To understand the definition and significance of cyber security • To analyze the evolution and historical context • To Identify the types of cyber threats and the current threat landscape • To learn the key principles of cyber security (CIA Triad, defense-in-depth) • To understand risk management strategies and legal/ethical considerations in cyber security	Unit 1: Introduction to Cyber Security (5 Hours) • Overview of Cyber security: Introduction to the field, its importance, and its role in protecting digital environments • Definition and Significance of Cyber security: Why cyber security matters in today's world of connected systems • Evolution and Historical Context: The development of cyber security from its early days to modern-day challenges • Cyber Threat Landscape: The nature of cyber threats, how they have evolved, and their impact • Types of Cyber Threats: Malware, phishing, ransomware, etc., and their impact on organizations • Key Principles of Cyber security: Confidentiality, Integrity, Availability (CIA Triad) and defense-in-depth

	• Risk Management in Cyber security: Identifying, assessing, and mitigating cyber security risks • Legal and Ethical Considerations: Overview of laws and regulations in cyber security; ethical responsibilities
• To understand fundamental networking concepts and protocols • To identify and analyze common network attacks • To learn about network security technologies like firewalls, IDS/IPS, and VPNs • To explore wireless network security risks and best practices • To implement security measures for securing network devices	Unit Two: Basics of Networking and Security (8 Hours) • Networking Fundamentals: Introduction to networking concepts and key networking protocols • Basics of TCP/IP and Network Protocols: Detailed overview of TCP/IP model, addressing, and common protocols • Common Network Attacks: Eavesdropping, man-in-the-middle, DoS attacks, and real-world examples • Network Security Technologies: Firewalls, intrusion detection/prevention systems (IDS/IPS), and Virtual Private Networks (VPNs) • Wireless Network Security: Risks and threats associated with wireless networks; securing Wi-Fi networks • Securing Network Devices: Best practices for securing routers, switches, and other network devices • Implementing Access Controls and Monitoring: Techniques to manage access and monitor network traffic
• To understand key security features of operating systems • To implement user account management and access controls • To learn about patch management and its importance • To apply antivirus and anti-malware protection strategies. • To understand encryption and secure boot processes for OS security	Unit Three: Operating System Security [8 Hours] • Basics of Operating System Security: Introduction to operating system security features • User Account Management and Access Controls: Managing user permissions and access controls • Patch Management: The importance of software updates and strategies for effective patch management • Antivirus and Anti-malware Protection: Role of antivirus software in cyber security; how to evaluate and select antivirus solutions

	• Encryption and Secure Boot: Techniques for securing data using encryption and ensuring a secure boot process • Endpoint Security: Best practices for endpoint security in operating systems
• To understand web application security and common vulnerabilities • To learn best practices for secure coding and secure browsing • To implement HTTPS, SSL/TLS for encrypted web communications • To conduct web security assessments using security tools like OWASP ZAP • To develop web security policies and understand compliance standards	Unit Four: Web Security [5 Hours] • Web Application Security Basics: Introduction to common web application vulnerabilities and their impact • Common Vulnerabilities in Web Applications: Analysis of SQL injection, cross-site scripting (XSS), and other vulnerabilities • Best Practices for Secure Coding: Guidelines for writing secure code and preventing common vulnerabilities • Secure Web Browsing: Safe browsing habits, avoiding phishing attacks, and recognizing threats • HTTPS and SSL/TLS: Importance of encrypted communication on the web and how to configure SSL/TLS for websites • Web Security Tools and Testing: Introduction to web security tools like OWASP ZAP and penetration testing practices • Web Security Policies and Compliance: Developing security policies and complying with industry standards (e.g., PCI DSS)
• To understand the importance of cyber security education and training • To develop incident response plans and conduct exercises • To explore cloud security fundamentals and best practices • To learn about threat intelligence and its role in cyber security • To learn emerging trends and continuously adapt to evolving threats	Unit 5: Best Practices in Security and Emerging Trends [4 hours] • Security Awareness and Training: Importance of cyber security education and creating a security-aware culture • Incident Response and Management: Developing incident response plans and conducting simulated exercises • Cloud Security Fundamentals: Security considerations in cloud environments and the shared responsibility model • Threat Intelligence and Information Sharing: The role of threat intelligence in cyber security and information-sharing communities

	• Future Trends in Cyber security: Emerging technologies, challenges, and the need for continuous learning
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Major Teaching and Learning Strategy:

The course employs two groups of teaching strategies. The first group consists of general strategies that can be applied to most units. These include providing reading materials for each topic, using lectures, facilitating group discussions, incorporating multimedia projectors, and engaging in brainstorming sessions. The second group focuses on specialized teaching methods tailored to specific units. A key strategy in this group is demonstrations, which will be used to provide hands-on experience and practical learning whenever possible.

Assessment Plan

a. In-semester (50 marks)

- i. Internal – I
- ii. Internal – II
- iii. Assignments
- iv. Practical Exam/ Viva /Case-studies

b. End-semester (50 marks)

- i. MCQ – 10
- ii. Subjective – 40

Laboratory (Not limited to):

- Network Scanning and Vulnerability Assessment: Perform network scanning to identify open ports and services, and assess vulnerabilities using scanning tools. Tools: Nmap, OpenVAS
- Malware Analysis and Detection: Analyze files to detect malware using both static and dynamic analysis techniques. Tools: Cuckoo Sandbox, ClamAV
- Firewall Configuration and Testing: Configure a firewall using iptables or pfSense, and test the firewall rules for security. Tools: iptables, pfSense, UFW.
- Web Application Security Testing: Test web applications for vulnerabilities like SQL injection and XSS, and learn how to secure applications against common attacks. Tools: OWASP ZAP, Burp Suite (community edition), DVWA
- Password Cracking and Hashing: Learn how to hash passwords and use cracking tools to break them, emphasizing strong password policies. Tools: John the Ripper, Hashcat
- Encryption and Decryption: Practice encrypting and decrypting data to understand how encryption secures sensitive information. Tools: GPG, OpenSSL
- IDS/IPS Configuration and Monitoring: Configure and monitor an Intrusion Detection or Prevention System to detect malicious activity on a network. Tools: Snort, Suricata
- Phishing Attack Simulation and Detection: Simulate phishing attacks and learn how to detect and respond to phishing emails. Tools: SET (Social-Engineer Toolkit), PhishTank
- Cloud Security Configuration: Configure and secure cloud infrastructure, ensuring best practices for security in cloud environments. Tools: OpenStack, Terraform, AWS Free Tier

- Incident Response Simulation: Simulate cyber security incidents such as data breaches or malware infections and practice incident response techniques to simulate and respond to cyber security incidents effectively. Tools: The Hive Project, Security Onion

Text and Reference Books:

- Computer Security Basics by Rick Lehtinen, Publisher: O'Reilly Media, 2nd edition (June 23, 2006), ISBN-10: 0596006691, ISBN-13: 978-0596006693, CBS PUBLISHERS & DISTRIBUTORS PVT. LTD, 01149347068
- Fundamentals of Computer Security by Josef Pieprzyk, Thomas Hardjono, Jennifer Seberry, Publisher: Springer, Softcover reprint of hardcover 1st ed. (December 1, 2010), ISBN-13: 978-3642077135, ISBN-10: 3642077137

Marking Scheme for Final Examination:

- Section A (MCQ): 10 Questions × 1 marks
- Section B (**Short Answer Question-SAQ**) – Total 7 Questions × 4 marks (Attempt any Six)
- Section C (**Long Answer Question-LAQ**) – Total 3 Questions × 8 marks (Attempt any Two)

Chapter	MCQ	SAQ	LAQ
Chapter 1	2	1	×
Chapter 2	2	2	1
Chapter 3	2	2	1
Chapter 4	2	1	1
Chapter 5	2	1	
10 MCQ Questions		Seven SAQ Questions	Three LAQ Questions

ITEX 325 Entrepreneurship

Course Title: Entrepreneurship

Code: ITEX 325

Credit Hours: 2 (1 TH + 1 PR)

Nature: Theory + Practical

Course Description and Goals

This course deals with software and software development methodologies. This course focuses on the knowledge of software engineering and its application in the real life for software development. It is concerned with theories, methods, and tools which are needed to develop computer software.

Learning Outcomes

On completion of this course, the students will be able to:

- Understand the key concepts and characteristics of entrepreneurship in the IT sector.
- Identify and evaluate business opportunities specific to the IT industry.
- Create a comprehensive business plan tailored to IT-based ventures.
- Demonstrate practical skills in launching and managing IT entrepreneurial initiatives.
- Analyze market trends, legal considerations, and ethical issues in IT entrepreneurship.

Content with specific objectives

Specific objectives	Contents
Understand entrepreneurship's core concepts, characteristics, and processes and highlight their relevance to the IT industry.	Unit 1: Introduction to Entrepreneurship and IT Ventures (5 Hours) • Definition and scope of entrepreneurship. • Characteristics of successful entrepreneurs. • Entrepreneurial motivations and decision-making processes. • Importance of entrepreneurship in economic and technological development. • IT entrepreneurship: unique challenges and opportunities. • Case studies of successful IT entrepreneurs and startups.
Enable students to recognize entrepreneurial opportunities and develop innovative business ideas.	Unit 2: Opportunity Identification and Business Idea Development (4 Hours) • Identifying and evaluating opportunities in IT markets. • Techniques for generating and refining business ideas. • Concept of feasibility study: Definitions and applications. • Importance of feasibility studies in entrepreneurial decision-making.

	● Feasibility analysis components: market, technical, organizational, and financial. ● Conducting feasibility analysis in the Nepali context with examples from the IT industry.
Develop skills to analyze industry dynamics and competitor landscapes.	Unit 3: Industry and Competitor Analysis (5 Hours) ● Industry trends analysis: tools and techniques. ● SWOT Analysis: Application of SWOT ● Identifying competitors and gathering competitive intelligence. ● Porter's Five Forces Model: application to IT industries. ● Industry types and emerging opportunities (e.g., SaaS, AI, IoT, Blockchain). ● Conducting a Competitive Grid Analysis to evaluate market positioning.
Equip students with skills to design, develop, and present business plans.	Unit 4: Developing a Business Plan (4 Hours) ● Business Plan. Importance and components of a business plan. ● Financial, marketing, and operational planning. ● Lean Canvas and Business Model Canvas frameworks. ● Techniques for pitching a business plan to investors and stakeholders.
Build an understanding of the legal, ethical, and societal aspects of entrepreneurship.	Unit 5: Legal, Ethical, and Social Dimensions in IT Ventures (5 Hours) ● Intellectual property rights (IPR) in IT: patents, copyrights, and trademarks. ● Data privacy, cybersecurity laws, and their implications. ● Corporate social responsibility (CSR) in IT businesses. ● Ethical dilemmas in tech entrepreneurship, including AI and user data.

Explore the practical challenges of launching, managing, and scaling IT ventures.	Unit 6: Launching, Managing, and Scaling IT Startups (7 Hours) - Strategies for raising capital: venture capital, angel investors, crowdfunding. - Building and managing high-performing IT teams. - Scaling Startups: operational and strategic considerations. - Managing day-to-day operations in tech startups. - Adapting to emerging trends and technologies.
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Major Teaching and Learning Strategy

- Interactive lectures with case studies from the IT sector.
- Workshops on business modeling and feasibility analysis.
- Guest lectures by experienced IT entrepreneurs.
- Group projects and peer discussions.
- Practical assignments to prototype and pitch IT business ideas.

Practical

- Conduct a market and competitor analysis using Porter's Five Forces.
- Develop a Lean Canvas for an IT startup.
- Prepare a comprehensive business plan for an IT venture and present it to the class.
- Simulate the launch of an IT product service or website.

Assessment Plan

- In-semester (50)**
Assignment, written exam, presentation, case study, report, objectives, attendance
- End-semester (50)**

Question Types	Marks Distribution	Total Question
MCQ	10 Marks	10 Questions
SAQ	24 Marks	7 Questions [Attempt 6 Questions]
LAQ	16 Marks	3 Questions [Attempt 2 Questions]

Marking Scheme for Final Examination:

- Section A (**MCQ**): 10 Questions × 1 marks
- Section B (**Short Answer Question-SAQ**) – Total 7 Questions × 4 marks (Attempt any Six)
- Section C (**Long Answer Question-LAQ**) – Total 3 Questions × 8 marks (Attempt any Two)

Learning Units	Total Credit Hours	MCQs	SQs	LQs	Remarks
Unit 1: Introduction to Entrepreneurship and IT Ventures	5	2	1		
Unit 2: Opportunity Identification and Feasibility Analysis	4	2	1	1	
Unit 3: Industry and Competitor Analysis	5	2	2	1	
Unit 4: Developing a Business Plan	4	1	1		
Unit 5: Legal, Ethical, and Social Dimensions in IT Ventures	5	1	1		
Unit 6: Launching, Managing, and Scaling IT Startups	7	2	1	1	

References:

- Osterwalder, A., & Pigneur, Y. *Business Model Generation*. Wiley.
- Blank, S., & Dorf, B. *The Startup Owner's Manual: The Step-by-Step Guide for Building a Great Company*. K&S Ranch.
- Ries, E. *The Lean Startup: How Today's Entrepreneurs Use Continuous Innovation to Create Radically Successful Businesses*. Crown Business.
- Barringer, B.R., & Ireland, R.D. *Entrepreneurship: Successfully Launching New Ventures*. Pearson.
- Schaper, M., Volery, T., Weber, P., & Lewis, K. *Entrepreneurship and Small Business*. Wiley.
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ITEX 326 IT and the Environment

Course Title: IT and the Environment

Course Code: ITEX 326

Credit Hours: 2 (1 Th + 1 Pr)

Nature: Theory and Practical

Course Description and Goals

This course explores the intricate relationship between Information Technology (IT) and the environment. It delves into the environmental impacts of IT, including energy consumption, electronic waste, and carbon emissions. Students will gain a comprehensive understanding of sustainable IT practices, green computing initiatives, and the role of IT in addressing global environmental challenges. The course also covers legal and ethical considerations related to IT and the environment.

Learning Outcomes

On completion of this course, the students will be able to:

- **Understand the Environmental Impact of IT:** Students will be able to identify and analyze the environmental impact of IT, including energy consumption, electronic waste, and carbon emissions.
- **Promote Sustainable IT Practices:** Students will be able to implement sustainable IT practices, such as energy efficiency, waste reduction, and responsible disposal of electronic waste.
- **Evaluate Green Computing Initiatives:** Students will be able to evaluate green computing initiatives, international standards, and certifications.
- **Apply Green IT Principles:** Students will be able to apply green IT principles to real-world scenarios, such as optimizing energy consumption, reducing waste, and promoting sustainable IT practices in organizations.
- **Promote Environmental Awareness and Responsibility:** Students will be able to raise awareness of environmental issues related to IT and promote responsible IT practices.

Content with specific objectives

Specific objectives	Contents
• Define sustainable IT and its importance • Identify the environmental impact of IT • Explore the role of IT in addressing global environmental challenges	Module 1: Introduction to IT and the Environment (6 hours) Overview on components of Environment (Natural, Human, and Built Environment), The triple bottom line: economic, social, and environmental sustainability, Introduction to Waste Electrical and Electronic Equipment (WEEE) or e-waste Concept of Sustainable IT, Digital revolution and its environmental footprint, The role of IT in climate change mitigation and adaptation and to support sustainable development goals initiatives, Smart cities and sustainable urban development

• Understand the StEP principles of Global initiatives. • Explain the Functions and campaigns of Basel Action Network. • Know about the decisions of European Union • Understand the current policy and steps taken by some Asian countries for e-waste management. • Learn about task forces for e-waste management.	Module 2: Environmental Issues, Initiatives and Standards (4 hours) Complexity of different IT equipment (composition) and issues related with toxins, precious metals, and other waste, power consumption and heat Global initiatives: United Nation (StEP principle), Basel Action Network, Basel Convention; EU: WEEE directive & RoHS, Current standard practices and adoption of different policy in some Asian countries,
• To understand the power problems and monitor its usage. • To analyze the power consumption issue & offer recommendations to reduce it. • To synthesize the cooling issues in the datacenter as well as optimize cooling cost.	Module 3: Minimizing Power Usage and Cooling (6 hours) Power problems, monitoring power usage, reducing power use, low-power computers, components Cooling costs, reducing cooling costs, optimizing air flow, adding cooling, data center design
• To understand and implement the process reengineering with Green in Mind. • To analyze and apply the steps for water conservation, recycling of organizational waste products such as paper, glass, plastic, to reduce energy consumption, the sources of pollutants and in what way it can be managed. • To define paper problems and its impact on environment. • To specify the profits of going paperless for organization, reducing cost and its effects on environments. • To define and explain the organizational approach for going paperless.	Module 4: Changing the way we work (4 hours) Rethinking Old Behaviors, Concept and Steps in optimizing Water & Energy usage, Pollutants and Recycling of office resources, Teleworkers and outsourcing, Paper problems, Paper and your Office, Going paperless, Intranets

- To understand the different recycling scenario and analyze on best practices. - To identify and implement energy-efficient solutions for office equipment and hardware, aiming to reduce overall power consumption and operational costs.	Module 5: Recycling and Hardware Considerations (6 hours) Current scenario, Means of Disposal, Life cycle, Extended Producer Responsibilities and Green Design Recycling companies, Hard Drive recycling, CDs and DVDs Certification programs, Energy stars, Servers, Hardware considerations, Remote Desktop
- To track and analyze greening of information systems using different metrics. - To understand the ways of improving technological infrastructure in ecofriendly way. - To learn how to keep the greening process alive even after achieving our goals. - To conduct audits and recommend improvements.	Module 6: Greening your Information System and Staying Green (6 hours) Initial Improvement Calculations, Change Business processes, Improve Technology Infrastructure Organizational Checkups, Equipment Checkups, Certifications, Helpful Organizations

Major Teaching and Learning Strategy (List of some Teaching and Learning Strategies for the faculty)

- Interactive Lectures
- Different Case studies
- Group discussions
- Project work and Presentations

Practical

- Material Flow Analysis of selected Electrical and Electronic Equipment of any Office or House. *(Related to Module 1)*
- Identify any real world environmental problem in your city and present a monitoring and mitigation plan through IT intervention. *(Related to Module 2)*
- Present a real world case study (any office) and ways to implement the process reengineering with Green in Mind. *(Related to Module 3 and 4)*
- Write a term paper on current scenario of e-waste management in any country. *(Related to Module 5)*

Assessment Plan

- In-semester (50)**
Assignment, written exam, presentation, case study, report, objectives, attendance
- End-semester (50)**

Question Types	Marks Distribution	Total Question
MCQ	10 Marks	10 Questions
SAQ	24 Marks	7 Questions [Attempt 6 Questions]
LAQ	16 Marks	3 Questions [Attempt 2 Questions]

Marking Scheme for Final Examination:

- Section A (**MCQ**): 10 Questions × 1 marks
- Section B (**Short Answer Question-SAQ**) – Total 7 Questions × 4 marks (Attempt any Six)
- Section C (**Long Answer Question-LAQ**) – Total 3 Questions × 8 marks (Attempt any Two)

Learning Units	Total Credit Hours	MCQs	SQs	LQs	Remarks
Module 1	6	2	1	1*	
Module 2	4	1	1		
Module 3	6	2	1	1*	
Module 4	4	1	1		
Module 5	6	2	1	1*	
Module 6	6	2	1	1*	

* LQs will be asked from any of these modules.

Additional one SQ can be asked from any of the modules.

References:

- Velte, T., Velte, A., Elsenpeter, R. C. (2008). Green IT: Reduce Your Information System's Environmental Impact While Adding to the Bottom Line: Reduce Your Information System's Environmental Impact While Adding to the Bottom Line. United Kingdom: McGraw Hill LLC.
- Lamb, J. (2009). The Greening of IT: How Companies Can Make a Difference for the Environment. India: Pearson Education.
- Smith, B. E. (2013). Green Computing: Tools and Techniques for Saving Energy, Money, and Resources. United Kingdom: CRC Press.
- Sharma, N., Verma, J. P., Gautam, S., Balas, V. E., & Krishnan, S. (2024). Green computing for sustainable smart cities: A Data Analytics Applications Perspective. CRC Press.